Computer Vision Approaches for Laboratory Vial Analysis

Recipe identification and mixture composition estimation from vial images

Prepared from Training model 2 and training model 3 notebook exports

Abstract

Automated interpretation of laboratory mixtures from images could support process monitoring, quality control, and error detection. This study investigated whether vial images contain sufficient visual information for two related tasks: qualitative identification of recipe identity and quantitative estimation of mixture composition. A pretrained ResNet18 was first fine-tuned to classify ten recipe categories. After records with missing recipe labels were removed, the model achieved 100.00% training accuracy, 97.37% validation accuracy, and 100.00% test accuracy on a random frame-level split. The task was then reformulated as three-output regression using actual recorded component volumes normalized to r100, r200, and r400. Under vial-held-out three-fold validation, the full-image regression model achieved mean MAE 0.1978 ± 0.0354 and mean R^2 −0.0646 ± 0.2998. A tight vial crop and a linear model based on six RGB statistics did not improve generalization. Grad-CAM suggested that the convolutional model used both the liquid region and contextual information from the vial and filling apparatus. The results show that recipe-associated visual patterns are readily distinguishable under the recorded conditions, whereas accurate composition estimation on unseen vials remains unresolved. More independent data, compositional output constraints, and vial- or session-level validation are required before deployment.

1 Introduction

Visual inspection is commonly used in laboratory and manufacturing settings to assess liquid color, transparency, fill level, and apparent composition. A reliable computer vision system could reduce operator variability and enable continuous monitoring. This study treated recipe identification and composition estimation as two stages of one investigation rather than as separate projects. The first stage asked whether known recipe categories could be recognized from their visual signatures. The second asked whether the same images supported direct estimation of the underlying component proportions.

Research Questions

- Can a transfer-learning model accurately identify the recipe associated with a vial image?
- Can a regression model estimate the actual relative volumes of three mixture components from an image?
- Do tight cropping or simple color statistics improve quantitative generalization to unseen vials?
- Which regions of the image influence the regression model's predictions?

2 Dataset Development

The analysis combined recipe metadata, frame-level labels, and image files extracted from a ZIP archive. Filename-based one-to-one matching confirmed that all 845 labeled frame records had corresponding images. The recipe table contained ten recipe IDs, numbered 5 through 14, with three vials per recipe and a mixture type such as Triple or a two-component pair.

Stage	Records	Purpose
Initial frame labels	845	All labeled frames matched to image files.
Classification data after removing missing recipe IDs	509	Excluded 336 between-fill or otherwise unlabeled frames.
Regression data after filtering and temporal selection	300	Retained in-fill frames and the final 10 frames for each recipe-vial combination.
Regression fold size	200 train / 100 test	One of three vials per recipe was held out in each fold.

Regression Targets

Missing component-volume entries were set to zero where the component was absent. The targets were then normalized by total volume: r100 = t100_ml / total_ml, r200 = t200_ml / total_ml, and r400 = t400_ml / total_ml. A validation check showed that the three ratios summed to exactly 1.0 for all 300 selected records. These targets represent actual recorded volumes, not recipe category labels.

Evaluation Design

The classification notebook used a randomized 70/15/15 frame-level split. This is useful as an initial model-development check, but consecutive frames from the same filling run can occur in different subsets and can be nearly identical. The regression notebook addressed this weakness by holding out one entire vial for every recipe in each fold. Each fold therefore contained all ten recipes in both training and testing, while the physical vials were disjoint.

3 Experimental Methods

Model	Output	Training setup
Recipe classifier	10 recipe classes	Pretrained ResNet18, cross-entropy loss, Adam optimizer at 1×10^{-4}, 10 epochs, batch size 32.
Full-image regressor	Three mixture ratios	Pretrained ResNet18 with a 512-to-3 final layer, MSE loss, Adam at 1×10^{-4}, 10 epochs, batch size 16.
Tight-crop regressor	Three mixture ratios	Same regression model applied to an 80×145 pixel vial crop resized to 224×224.
Color baseline	Three mixture ratios	Linear regression using mean and standard deviation of the red, green, and blue channels from the crop.

4 Results

4.1 Recipe Identification

After removal of records with missing recipe IDs, the classifier was trained on 356 images, validated on 76, and tested on 77. Training loss decreased from 2.0456 to 0.0727 over ten epochs. The executed outputs reported the following results:

Measure	Result
Training accuracy	100.00%
Validation accuracy	97.37%
Test accuracy	100.00%
Test errors	0 of 77

The confusion matrix was perfectly diagonal across the ten recipe classes. This shows that the network learned highly discriminative visual signatures in the available frame-level dataset. However, the random frame split likely gives an optimistic estimate of deployment performance because frames from a single dispensing sequence share the same vial, apparatus state, lighting, and near-identical temporal context.

4.2 Full Image Composition Regression

A first single-fold run achieved MAE 0.1827 and R^2 0.1453. The complete three-fold rerun is the more representative result because every vial served as the held-out vial once:

Fold	MAE	R^2
1	0.1910	−0.0391
2	0.1583	0.2892
3	0.2442	−0.4439
Mean ± SD	0.1978 ± 0.0354	−0.0646 ± 0.2998

The large fold-to-fold variation indicates sensitivity to the physical vial or acquisition conditions. The negative mean R^2 means that, averaged across folds, the predictions did not improve on a constant mean-target predictor. The MAE corresponds to an average absolute ratio error of about 0.20 per component.

4.3 Preprocessing and Baseline Experiments

Approach	Observed result	Interpretation
Tight-crop ResNet18, fold 1	MAE 0.2401; R^2 −0.3537	Cropping to the narrow vial region degraded performance relative to the full image.
RGB linear baseline, fold 1	MAE 0.2472; R^2 −0.3684	Six global color statistics were insufficient on the executed split.
RGB linear baseline, folds 1–3	MAE 0.2472, 0.1988, 0.2174; R^2 −0.3684, 0.0533, −0.2158	Performance was inconsistent and average explanatory power was negative.

5 Model Interpretation

Grad-CAM was applied to the final convolutional block of the regression ResNet18. For images with visible liquid, the activation map concentrated on the liquid region and the lower vial. When little or no liquid was visible, attention shifted toward the vial and surrounding filling apparatus. This supports two conclusions: the model was using physically plausible liquid information, but it also depended on contextual cues outside the tight crop.

That behavior is consistent with the crop experiment. Removing the apparatus, vial outline, fill geometry, and broader scene removed predictive context as well as nuisance information. A better preprocessing strategy should isolate a moderately expanded region of interest or segment the vial and liquid while preserving fill-height and nearby geometric cues.

6 Discussion

Why Classification Was Easier

Recipe classification only required choosing among ten known mixture identities. Because each recipe has a fixed composition and was recorded in repeated frames, the network could exploit any stable cue correlated with recipe identity. Regression required estimating three continuous proportions and was evaluated on different physical vials. It therefore exposed variation that the classifier's random frame split did not test.

What the Regression Metrics Show

- Training loss fell sharply, so optimization succeeded; the limitation is generalization, not failure to fit the training data.
- Fold 2 achieved positive R^2, showing that usable visual signal exists under some held-out conditions.
- Negative R^2 in folds 1 and 3 indicates that vial-specific or session-specific differences can overwhelm the composition signal.
- The tight crop and RGB baseline results show that composition is not captured reliably by a very small region or six global color statistics alone.
- The three predicted ratios were produced by an unconstrained linear output layer, so predictions were not guaranteed to be nonnegative or sum to one.

Data Integrity and Reporting Notes

Executed cell outputs were treated as the authoritative numerical record. Some notebook narrative cells contain older values that do not match adjacent outputs. In particular, an earlier classification summary reports different accuracies, and an intermediate RGB paragraph reports MAE 0.1450 and R^2 0.1849; that RGB result is not reproduced by the later executed vial-held-out cells. It is therefore treated as a preliminary observation rather than a validated result. The final Grad-CAM interpretation is qualitative because the notebook contains an earlier undefined target reference before the later RegressionOutputTarget implementation.

7 Limitations

- Only three physical vials were available per recipe, limiting the estimate of vial-to-vial variability.
- The classification experiment used a random frame-level split, so closely related frames may have appeared in different subsets.
- The images came from a limited acquisition setting and did not provide independent external recording sessions.
- Regression performance varied substantially across folds, and the aggregate dataset was small for end-to-end deep learning.
- The regression output layer did not enforce nonnegative ratios or the required sum-to-one constraint.

8 Recommendations

Priority	Action	Reason
1	Evaluate by independent dispensing run or recording session, with all frames from a vial confined to one split.	This is the most important control against temporal and vial-level leakage.
2	Collect more vials per recipe across lighting, camera position, apparatus state, and day.	Three vials per recipe cannot characterize real acquisition variability.

Priority	Action	Reason
3	Constrain outputs with softmax or another compositional model so ratios are nonnegative and sum to one.	The targets are compositions and the model should respect that geometry.
4	Use a wider region of interest, timestamp masking, or vial/liquid segmentation instead of the 80×145 crop.	Preserves fill-height and apparatus geometry while reducing irrelevant background.
5	Report per-component MAE and bias in addition to aggregate MAE and R^2.	Identifies whether t100, t200, or t400 is systematically harder to estimate.
6	Compare regularized tabular and image baselines, including color histograms, HSV/Lab features, fill height, and partial fine-tuning.	Separates the value of deep spatial features from simpler physical measurements.
7	Use validation-based checkpointing, learning-rate scheduling, and repeated seeded runs.	Reduces dependence on one stochastic training trajectory and supports reproducibility.

9 Conclusion

The notebooks document a sound progression from recipe recognition to composition estimation. The cleaned classifier showed that ResNet18 can recognize recipe-associated visual patterns in the recorded frames, but the perfect random-split test result is likely inflated by shared sequence context. The vial-held-out regression experiment provides the more credible measure of performance for error identification and composition estimation. Its mean MAE of 0.1978 and mean R^2 of −0.0646 establish a clear baseline: the images contain useful signal, but the current dataset and model do not yet generalize reliably across vials.

The next effective model should retain regression, enforce compositional constraints, expand independent data collection, and validate by vial or session. These changes target the primary failure mode revealed by the experiments rather than merely increasing network complexity.